

LALITH SAGAR KAMBALA

Austin, TX | (913)-488-0303 | lalithsagarkambala789@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

Full Stack & AI Software Engineer who builds with ownership from architecture decisions to production deployments. I gravitate toward hard problems at the intersection of AI and product, turning ambiguous requirements into reliable, scalable systems that real users depend on. Driven by outcomes over output, I care about why something is built as much as how.

EDUCATION

Master of Science: Computer Science

Aug 2023 to May 2025

University of Central Missouri - Lee's Summit, MO – **GPA:** 3.55/4.00

EXPERIENCE

Full Stack AI Developer

Jul 2026 to Present

Relativity, Inc. – TX, United States

- Engineer distributed, cloud-native public safety platforms across CAD, Dispatch, Jail Management, Court Systems, Incident Reporting, and GIS, designing full-stack services and workflows with React, TypeScript, MongoDB, and Neo4j.
- Build scalable data and service architectures for real-time cross-module communication, leveraging Neo4j graph databases for relationship-driven workflows and implementing JWT and AWS Cognito authentication and authorization for secure, multi-system access.
- Improve system scalability, availability, and traffic resilience across AWS-hosted microservices by designing service deployment patterns, configuring NGINX load balancing, and optimizing distributed application components for mission-critical workloads.

Founding Software Engineer

Sep 2025 to June 2026

Todaiyo.ai – MA, United States

- Led the end-to-end development of a 0→1 AI SaaS platform, architecting scalable microservices, backend infrastructure, APIs, and cloud deployments while owning the complete software lifecycle from product planning, technical design, implementation, testing, documentation, and production releases.
- Built AI-powered product capabilities using OpenAI LLMs, RAG, vector embeddings, and multimodal inference for text, voice, and image understanding, translating business requirements into intelligent workflows that improved customer experience and accelerated product growth.
- Drove product growth and business outcomes by launching subscription monetization with Stripe, implementing product analytics (GA4/GTM), optimizing user acquisition and conversion funnels, collaborating on product strategy, and delivering data-driven features that contributed to recurring revenue and customer adoption.

AI Full Stack Engineer (*Part-time*)

Sep 2025 to May 2026

Consumer Genie – United States

- Architected production-grade multi-agent AI systems using LangChain, LangGraph, MCP (Model Context Protocol), OpenAI, and Claude, implementing agent orchestration, context engineering, prompt optimization, memory management, and structured outputs for reliable autonomous reasoning.
- Engineered scalable LLM inference pipelines by optimizing latency, accuracy, and token efficiency through prompt engineering, model routing, fallback strategies, evaluation frameworks, and algorithmic reasoning workflows for complex multi-step decision making.
- Developed semantic retrieval and AI knowledge pipelines using vector embeddings, MongoDB, PostgreSQL, Redis, and Kafka, implementing RAG, context retrieval, data validation, and continuous monitoring to improve AI response quality, reliability, and production performance.

Full Stack Software Engineer

Jul 2025 to Oct 2025

IEJL – CA, United States

- Developed responsive web applications using Next.js, React, TypeScript, and Tailwind CSS, translating Figma designs into production-ready interfaces and integrating REST APIs across a microservices architecture.
- Built secure backend features with Supabase, OAuth2, JWT, and RBAC, optimizing database schemas and API workflows to improve scalability, security, and performance.
- Deployed containerized applications using Docker on AWS with CI/CD pipelines, while collaborating in Agile teams to reduce development cycles by 35% and improve system reliability by 30%.

Web Development Intern

Sep 2022 to Nov 2022

Coincent Technologies – India

- Developed secure backend integrations between EHR systems and healthcare platforms, improving appointment processing efficiency by 35% through optimized REST APIs and real-time data synchronization.
- Engineered scalable backend services and optimized databases, reducing data inconsistencies by 40% and improving healthcare system reliability.

CORE COMPETENCIES

- Languages & Frameworks: Python, TypeScript, JavaScript, React, Next.js, Node.js, FastAPI, Flask, Spring Boot, Tailwind CSS
- Backend & Distributed Systems: REST APIs, GraphQL, Microservices, System Design, Event-Driven Architecture, Kafka, WebSockets, Async Processing
- Databases: MongoDB, PostgreSQL, MySQL, Redis, Neo4j, Supabase, Pinecone, FAISS, Vector Search
- Real-Time & Event-Driven: Real-Time & Event-Driven: System Design, WebSockets, Kafka, event-driven microservices, async processing, distributed systems, relational databases, RAG Pipelines, Chat Orchestration, SEO, custom analytics/tracking
- API Integration: RESTful APIs, ERP/CRM systems, GraphQL, Redis, Stripe, In-App purchases(IAP), HubSpot, OpenAI, Claude API, Web Scraping (DDG/Serper), Tavily
- Design & UX: Figma (UI/UX design, wireframing, app screen design), responsive design, design systems, component libraries, UX design principles, Core Web Vitals
- Security: OAuth2, JWT, authentication, authorization, privacy, secure deployment
- AI/ LLM: LangChain, LangGraph, Embeddings, Fine-tuning, MCP, ML Inference, RAG pipelines, LLM-agnostic architecture, Agentic orchestration, Prompt Engineering
- Cloud & DevOps: AWS (EC2, RDS, S3, CloudFront, Cognito, CloudWatch), NGINX, Docker, Kubernetes, CI/CD, Git (code reviews), GCP, Jira, Agile/Scrum, Linux, Netlify, Vercel
- Testing & Observability: Unit/Integration testing, CloudWatch, Logging, Error Monitoring, GA4 Analytics, GTM

ACHIEVEMENTS & CERTIFICATIONS

- Anthropic - MCP, Agents & Sub-Agents (Advanced)
- Oracle Cloud Infrastructure Generative AI Professional (1Z0-1127-24)
- Introduction to Large Language Models (LLM) by Google Cloud
- Accenture North America Data Science and Visualization Job Simulation on Forage

PROJECTS

AI- Debugging Agent using ollama – Pinecone

Mar 2026

- Built a modular agentic debugging pipeline with CodeLlama fine-tuned via RAG (Pinecone), featuring retrieval orchestration, structured outputs, namespace-based multi-repo querying, and privacy-first local vector store mode for on-device repository data.
- Designed evaluation frameworks with multi-agent RFT techniques, measuring reasoning quality, inference latency, throughput, and reliability; calculated entropy differentials across model variants and implemented production-grade logging, LLM/Pinecone fallback strategies, and query tracking across production workloads.

Article Semantic Search using LLM's

Aug 2025

- Designed a financial news research platform leveraging LangChain's Unstructured URL Loader and OpenAI embeddings for precise content indexing and retrieval.
- Deployed FAISS-based semantic search with embedding vector storage, and built an interactive Streamlit app allowing users to process URLs and obtain contextually accurate answers from an LLM-driven pipeline.

Brain Tumor Detection Using Deep Learning

Jan 2023 - Apr 2023

- Constructed a CNN model with Conv2D, Batch Normalization, MaxPooling, Dropout, and Dense layers using ReLU activation to classify MRI scans with 94% accuracy, enhancing feature extraction and reducing overfitting by 15%.
- Automated early tumor detection workflows, cutting diagnosis time by up to 40% and improving patient care through scalable, real-time image analysis.

Blockchain project on Property Management System

Aug 2022 - Dec 2022

- Built **Block Wizz**, a decentralized blockchain-based tax payment solution that processed 500+ transactions with 100% transparency, ensuring tamper-proof records and full auditability across the network.
- Incorporated SHA-256 hashing to securely chain payment histories, reducing fraudulent receipts and duplicate payments by 95% while enabling real-time, verifiable transaction tracking.